Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 06: Python Programming II

Digital for Executive

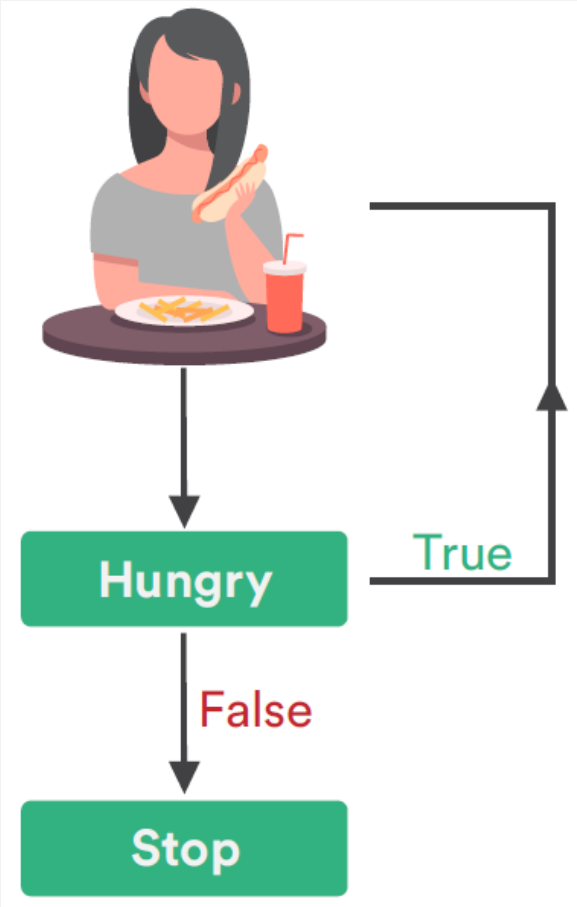
Outline

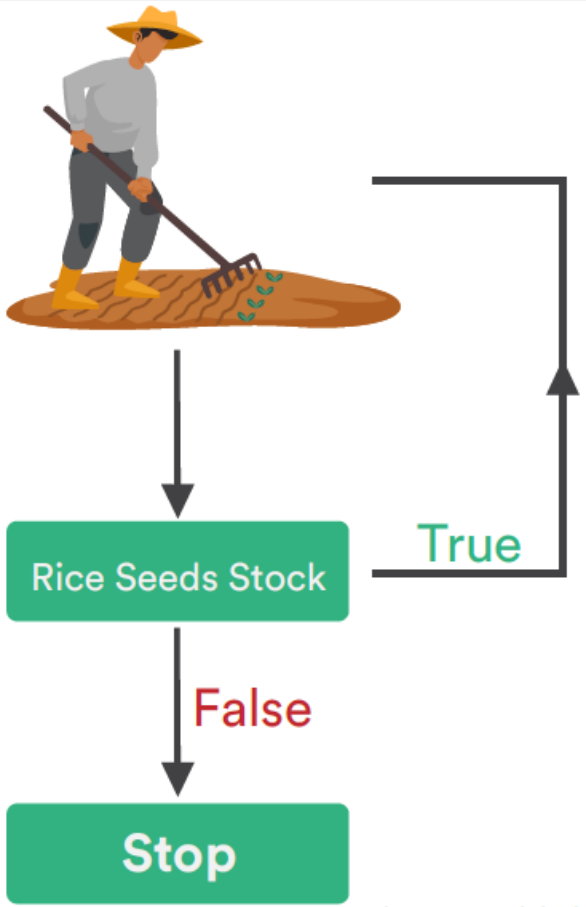
- Python Looping Statements
- Python Collection Data Types (Part 1)
- Exercise

Python Looping Statements

Loop Statements Introduction

Loop Statements Introduction

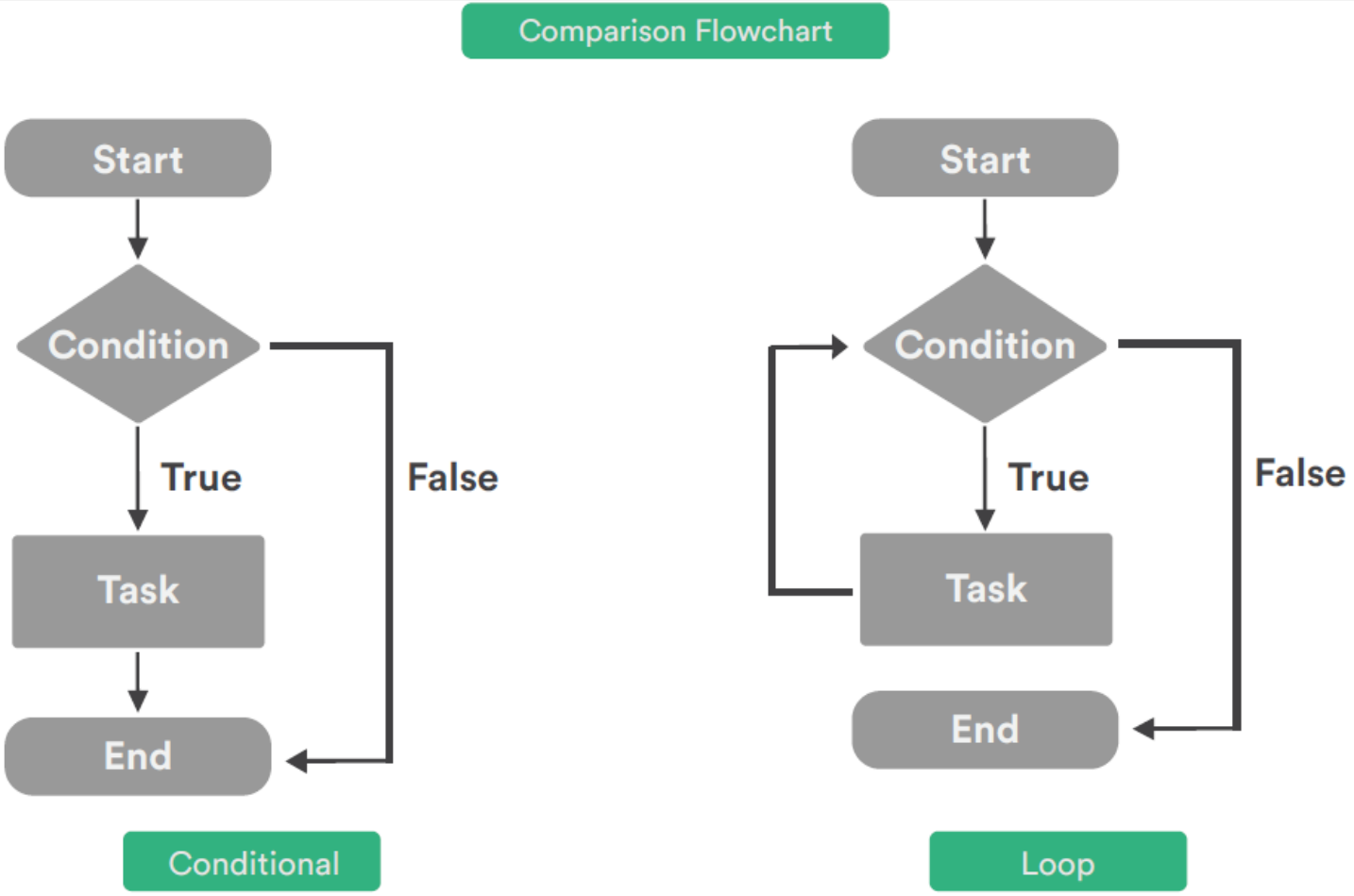




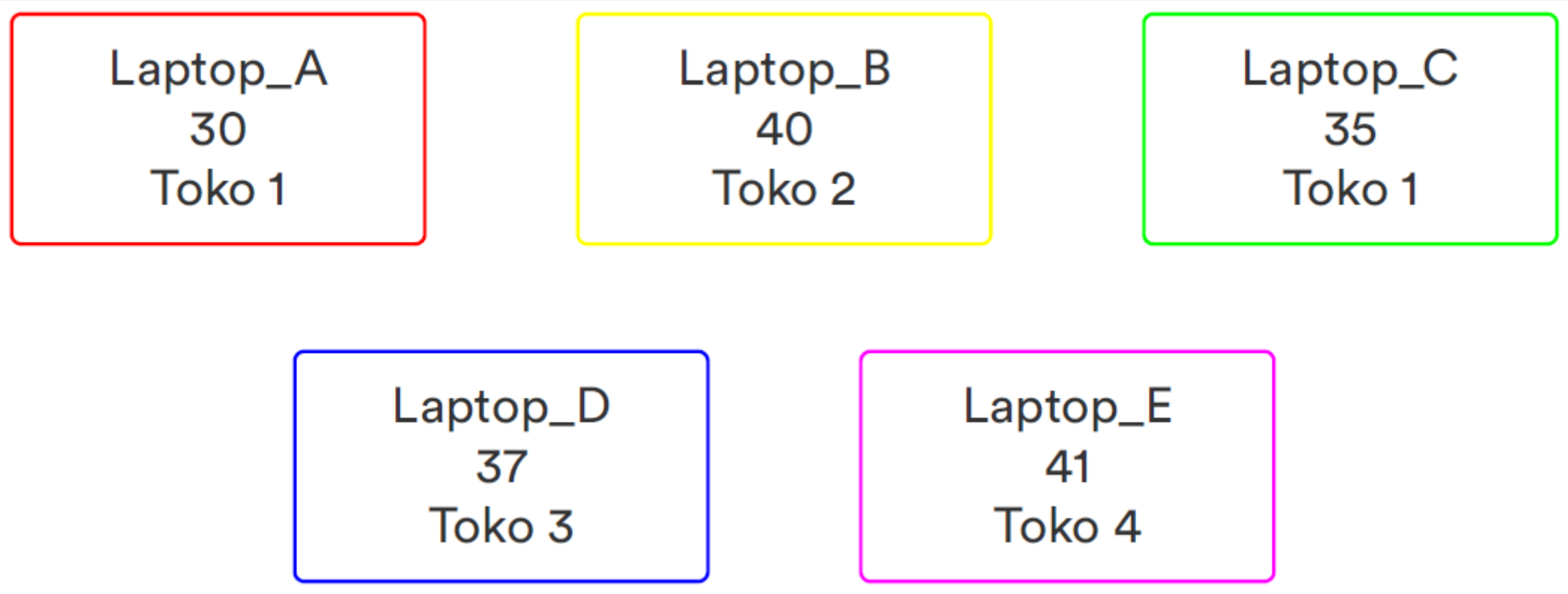
Loop Statements Introduction

Conditional Statement	Loop Statement
Need Boolean	Need Boolean
	Can Repeat Task

Loop Statements Introduction

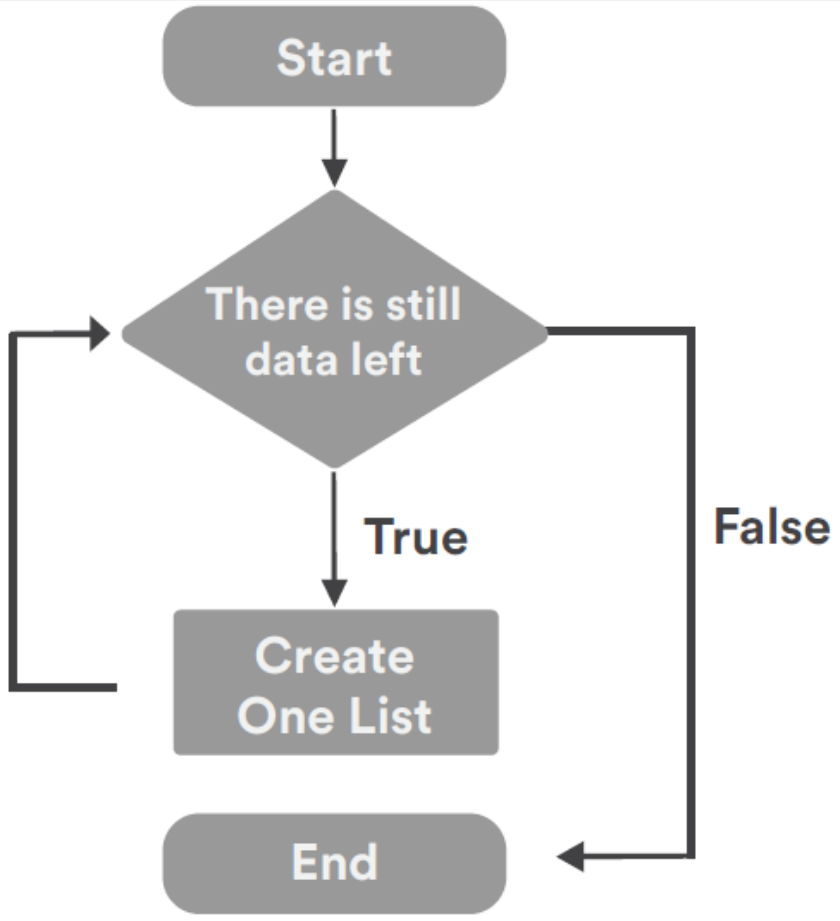


Loop Statements Introduction



Loop Statements Introduction

Loop Statements Introduction



Loop Statements Introduction

8.75 9

7.20 → 7

6.33 6

What, When, Why, How

What, When, Why, How

WHAT IS LOOP?

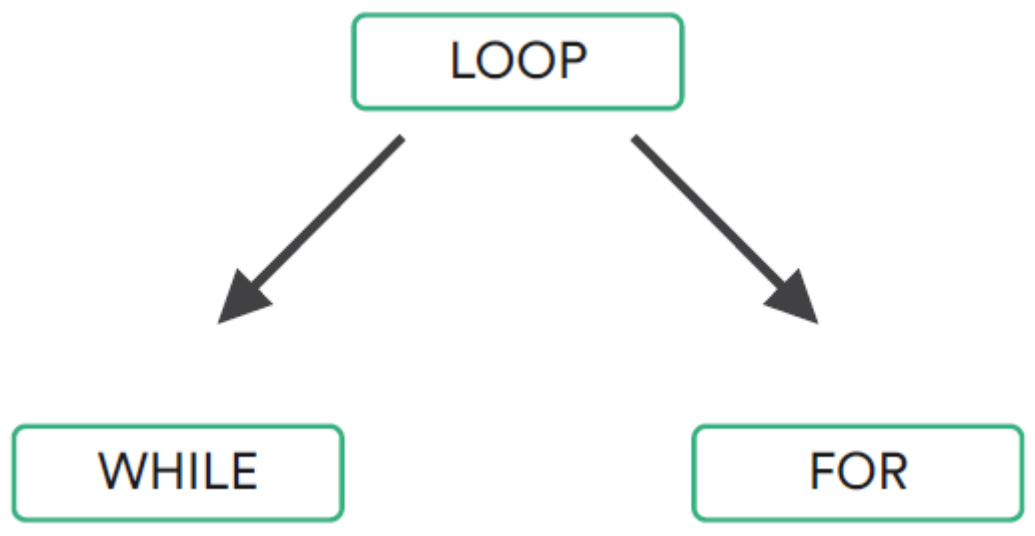
WHEN?

WHY?

HOW?

Two Types of Looping

Two Types of Looping



While Loop

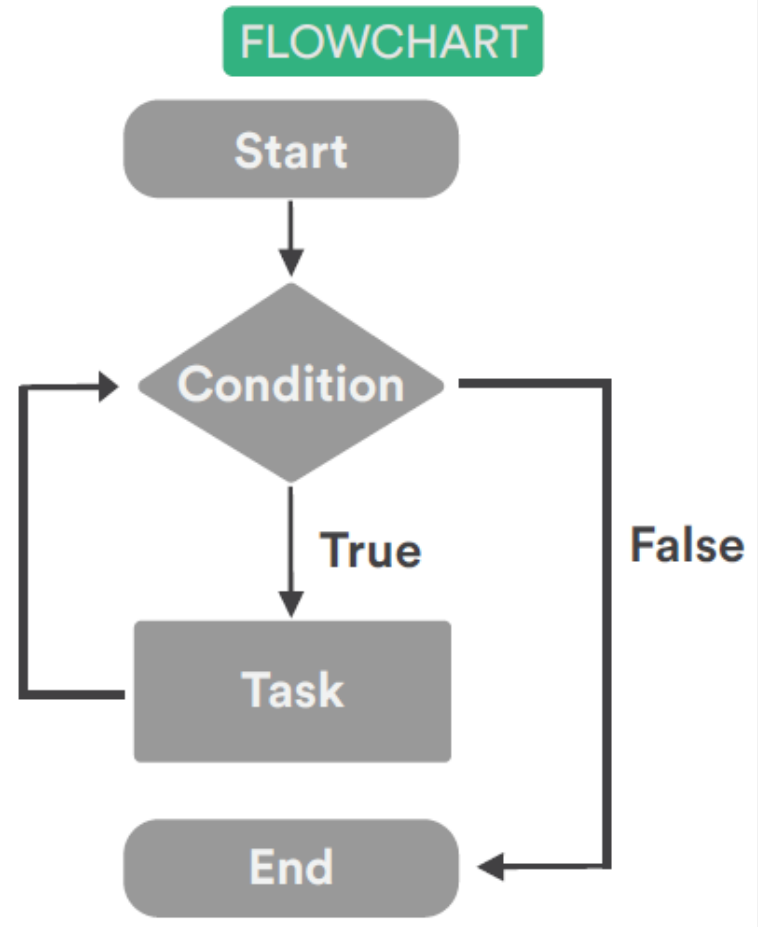
While Loop

SYNTAX

```
while condition :
    statement 1
    statement 2
    ...
    statement n
    compound statement
```

indentation

Executed statements While still True



For Loop

For Loop

SYNTAX

```
for val in range(start, stop, step):
    statement 1
    statement 2
    ...
    statement n
```

indentation

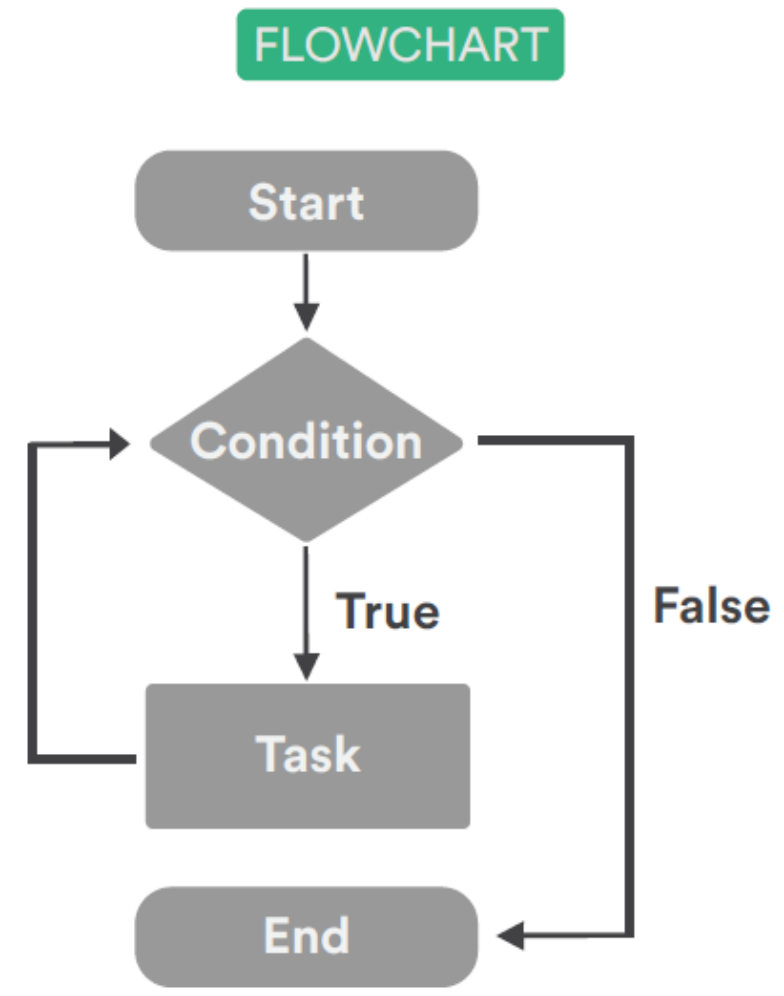
Executed statements 1 2 .. and n repeatedly

range() Function

range() Function mengembalikan urutan angka, dimulai dari 0 secara default, dan bertambah 1 (secara default), dan berhenti sebelum angka yang ditentukan.

```
range(start=0, stop, step=1)
```

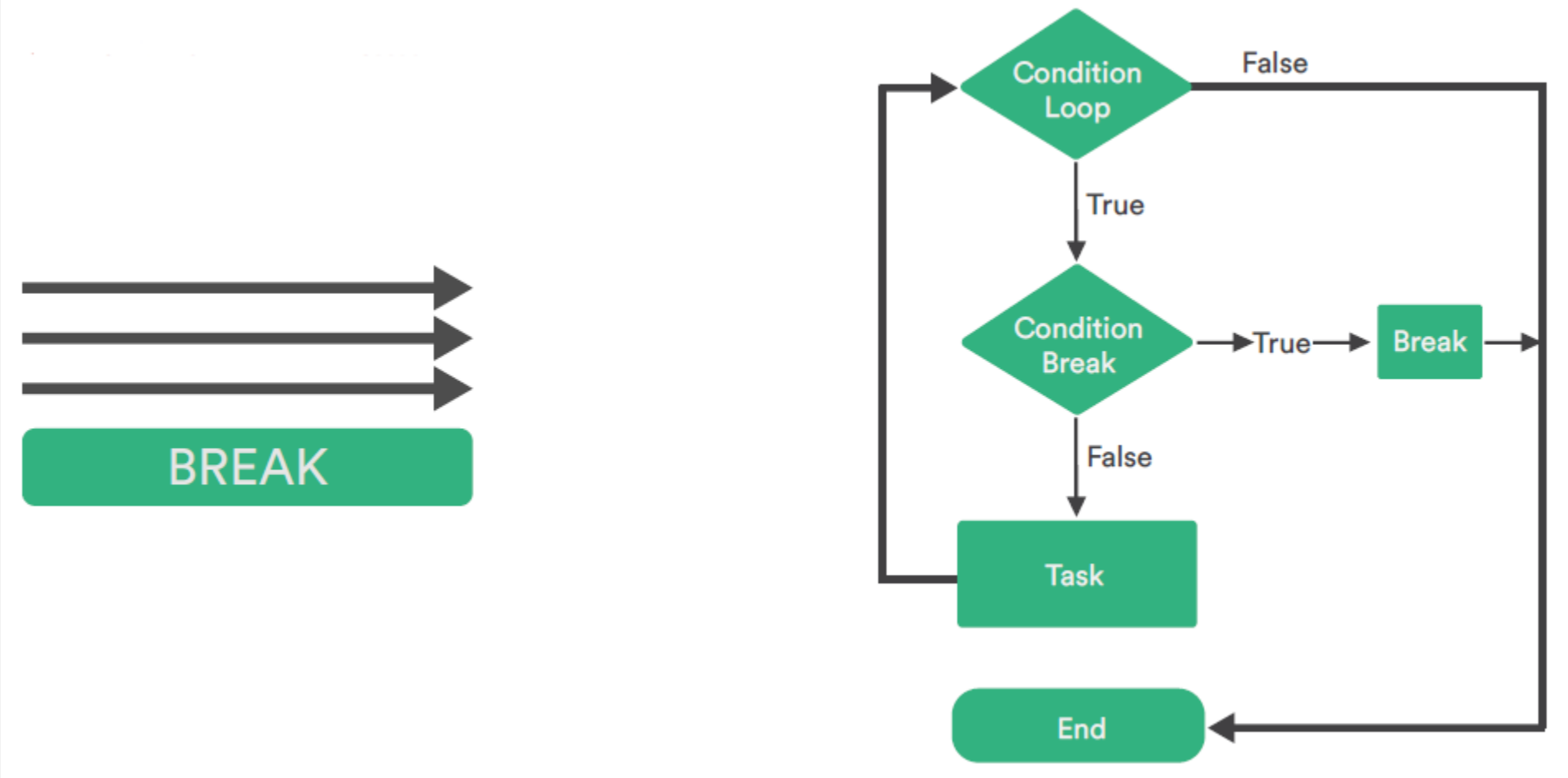
For Loop



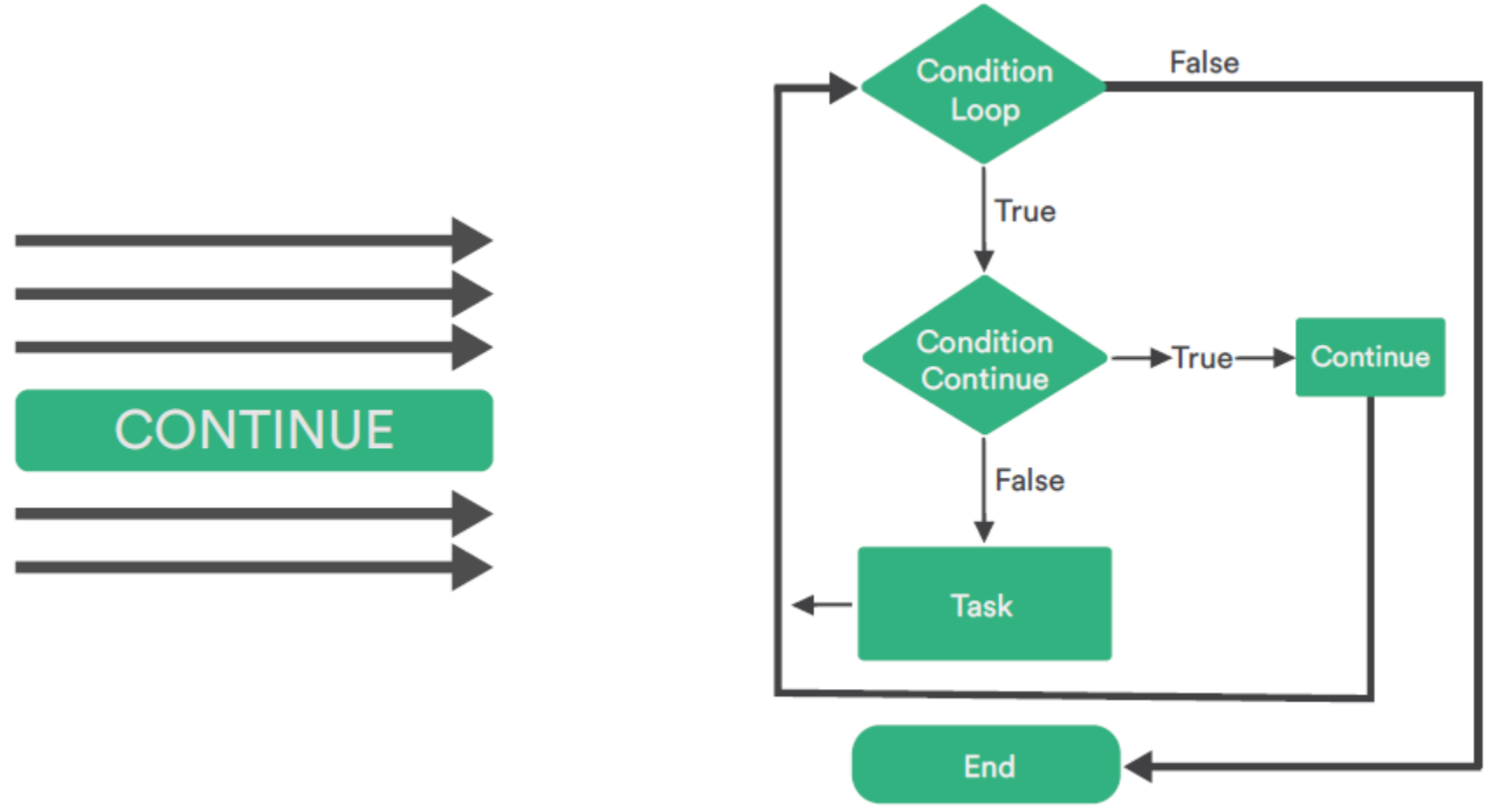
Break & Continue

Break & Continue

Break

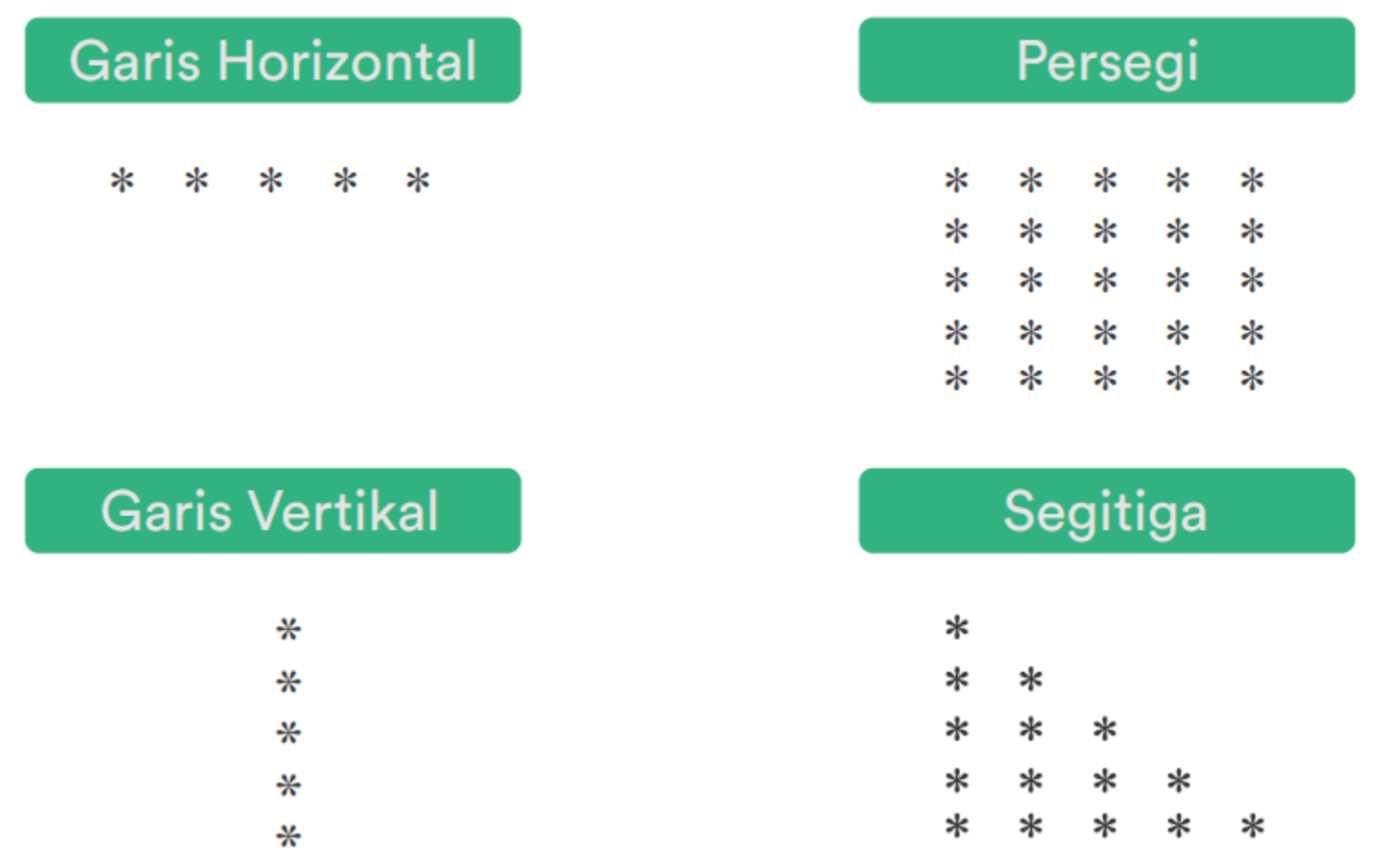

BREAK
Condition Loop
False
True
Condition Break
True
Break
False
Task
End

Continue


CONTINUE
Condition Loop
False
True
Condition Continue
True
Continue
False
Task
End

Loop Drawing

Loop Drawing



Loop Drawing

```
stars = ' '

for i in range(5) :
        stars += '* '

print(stars)

# result => * * * * *
```

Loop Drawing

```
stars = ' '

for i in range(5) :
    stars += '*\n'

print(stars)

# result => *
#           *
#           *
#           *
#           *
```

Loop Drawing

```
stars = ' '
size = 5

for i in range(size) :
    for j in range(size) :
        stars += '* '
    stars += '\n'

print(stars)

# result => * * * * *
#           * * * * *
#           * * * * *
#           * * * * *
#           * * * * *
```

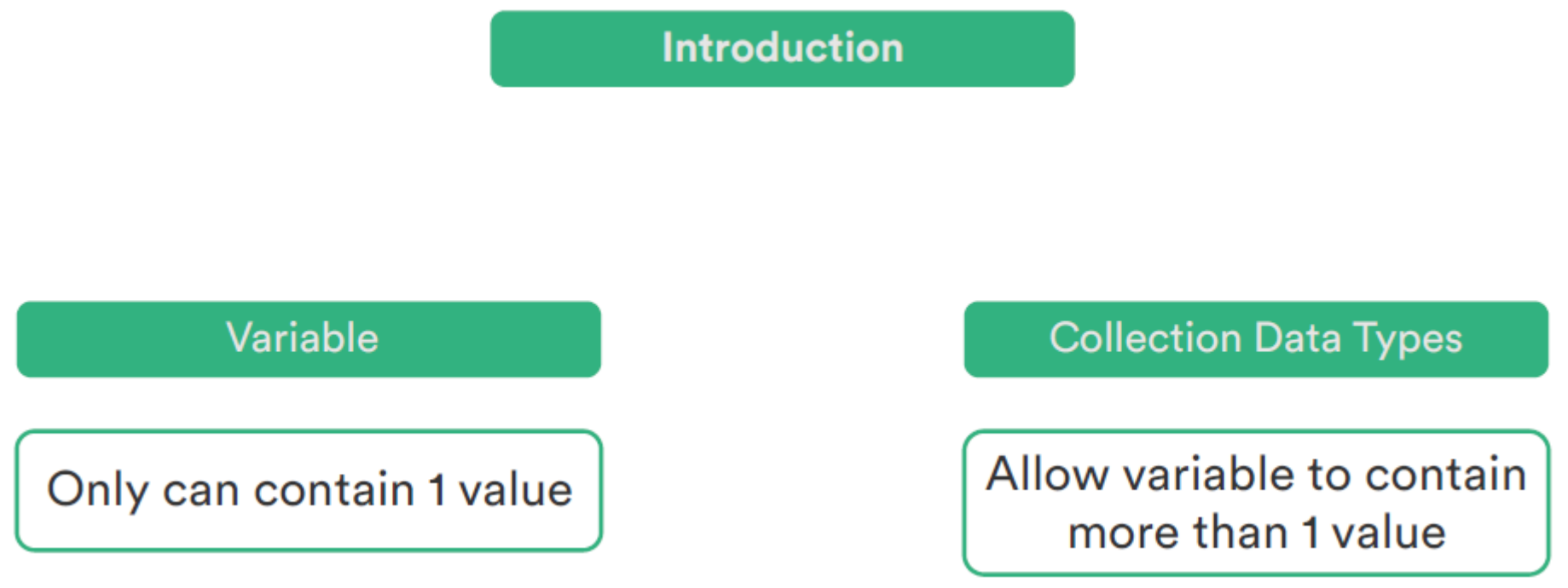
Loop Drawing

```
stars = ' '
size = 5

for i in range(size) :
    for j in range(1 + i) :
        stars += '* '
    stars += '\n'

print(stars)

# result => *
#           * *
#           * * *
#           * * * *
#           * * * * *
```

Python Collection Data Types (Part 1)

Collection Data Types Introduction

Collection Data Types Introduction



Collection Data Types Introduction

Python Collections

- List

```
listContoh = ['hello', 1, 1, 3, True]
print(listContoh,type(listContoh)) # ['hello', 1, 1, 3, True] <class 'list'>
```

- Tuple

```
tupleContoh = ('hello', 1, 1, 3, True)
print(tupleContoh,type(tupleContoh)) # ('hello', 1, 1, 3, True) <class 'tuple'>
```

- Set

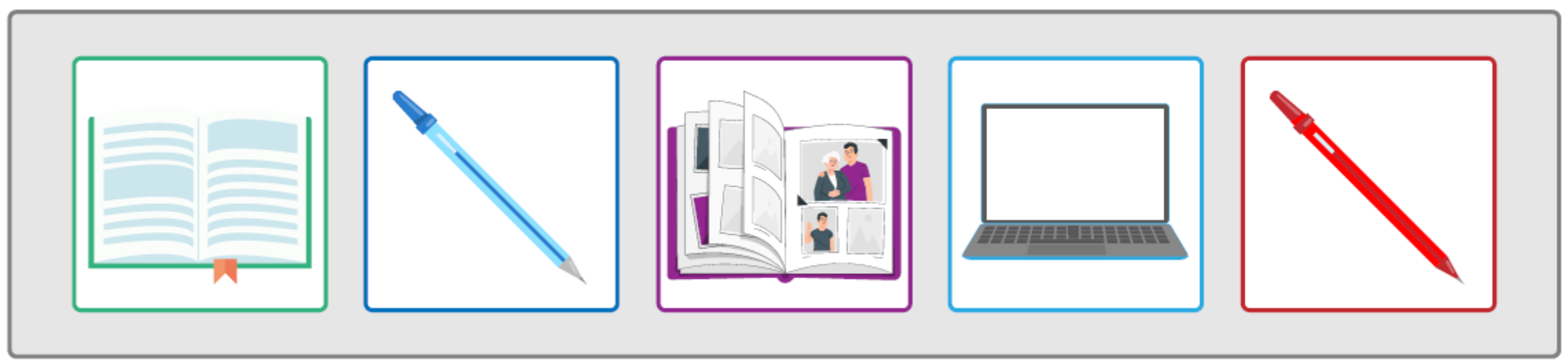
```
setContoh = {'hello', 1, 1, 3, True}
print(setContoh,type(setContoh)) # {1, 'hello', 3} <class 'set'>
```

- Dictionary

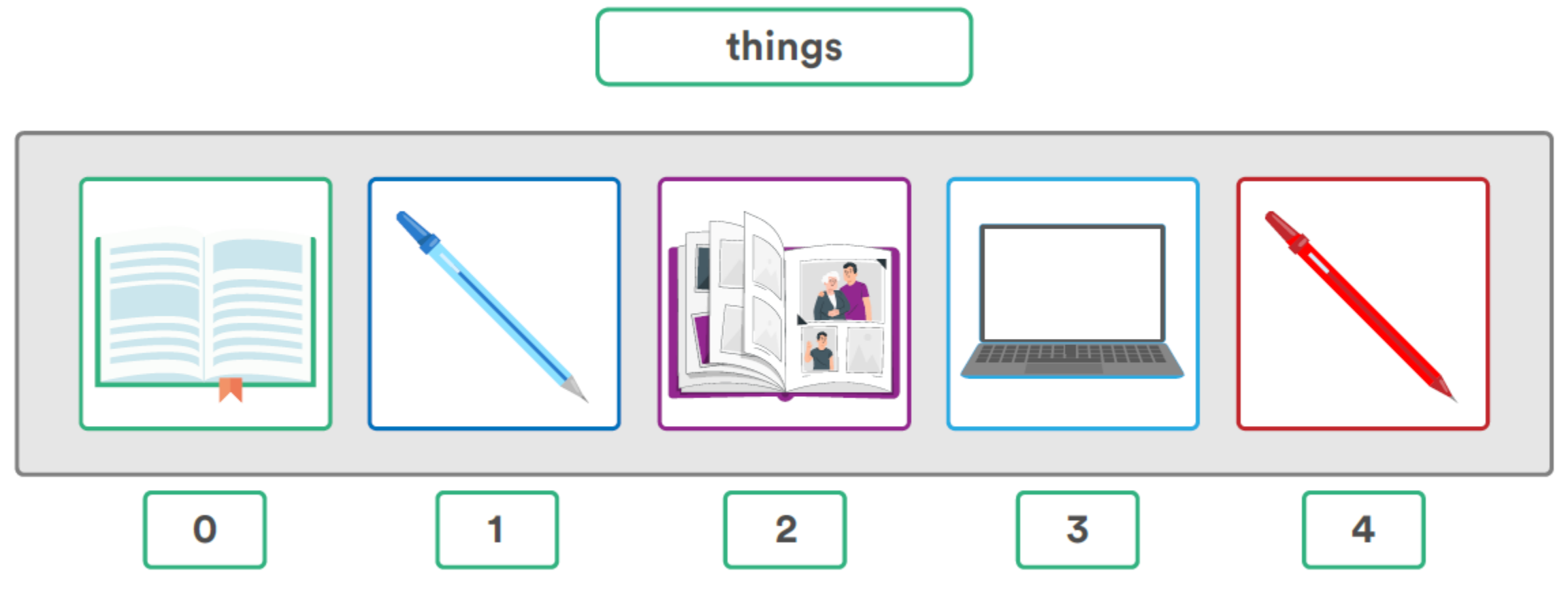
```
dictContoh = { 'nama' : 'Ricky', 'usia' : 25 }
print(dictContoh,type(dictContoh)) # { 'nama' : 'Ricky', 'usia' : 25 } <class 'dict'>
```

List

List Introduction



List Introduction



List Introduction

- It has a name, things
- divided into five parts
- every part have number
- save various of data type

List Introduction

Syntax List

Syntax List

```
# index =>          0        1   2   3   4
listContoh  =  ['hello',    1,  1,  3,  True]
```

Access Data

```
# index =>          0        1   2   3   4
listContoh  =  ['hello',    1,  1,  3,  True]

print(listContoh[0])   #  hello
print(listContoh[2])   #  1
print(listContoh[4])   #  True
```

Access Data Negative Indexing

```
# index =>          -5      -4  -3  -2  -1
listContoh  =  ['hello',    1,  1,  3, True]

print(listContoh[-1])  #  True
print(listContoh[-2])  #  3
print(listContoh[-5])  #  hello
```

Access Data Range of Indexes

```
# index =>        0/-5      1/-4    2/-3    3/-2    4/-1
listContoh  =  ['hello',      1,      1,      3,   True]

print(listContoh[-4 : -1])  #  [1, 1, 3]
print(listContoh[1 : 4])  #  [1, 1, 3]
print(listContoh[-5 : 2])  #  ['hello', 1]
print(listContoh[0 : -2])  #  ['hello', 1, 1]
print(listContoh[2 : ]) #  [1, 3, True]
```

Change Data

```
# index =>        0/-5      1/-4    2/-3    3/-2    4/-1
listContoh  =  ['hello',      1,      1,      3,   True]

listContoh[1]  =  'test'
listContoh[-2]  =  'coba'

print(listContoh)   #  ['hello', 'test', 1, 'coba', True]
```

Add Data

```
# index =>        0/-5      1/-4    2/-3    3/-2    4/-1
listContoh  =  ['hello',      1,      1,      3,   True]

listContoh.append ('coba')
listContoh.insert (1,'boleh')

print(listContoh)   #  ['hello', 'boleh', 1, 1, 3, True, 'coba']
```

Delete Data

```
# index =>        0/-5       1/-4       2/-3    3/-2    4/-1
listContoh  =  ['hello',    'coba',    'coba',    3,    True]

listContoh.remove('coba') # remove first coba
listContoh.pop() # remove last item
del listContoh[1]  # remove item at index 1

print(listContoh)   #  ['hello', 3]

listContoh.clear() #  remove all items

print(listContoh)   #  [ ]
```

Loop Through a List

```
# index =>        0/-5      1/-4    2/-3    3/-2    4/-1
listContoh  =  ['hello',      1,      1,      3,   True]

for item in listContoh :
    print(item)

# hello
# 1
# 1
# 3
# True
```

Check if an Item Exists in a List

```
# index =>        0/-5      1/-4    2/-3    3/-2    4/-1
listContoh  =  ['hello',      1,      1,      3,   True]

if 'hello' in listContoh :
    print("Ada string hello di listContoh")
else :
    print('string hello tidak ditemukan di listContoh')

# Ada string hello di listContoh
```

Length of List

```
# index =>        0/-5      1/-4    2/-3    3/-2    4/-1
listContoh  =  ['hello',      1,      1,      3,   True]

print(len(listContoh))  #  5
```

Copy a List

```
# index =>      0/-5      1/-4      2/-3    3/-2    4/-1
listContoh  =  ['hello',   'coba',   'coba',    3,   True]

newList1 = listContoh
newList1[1] = 'test'

newList2 = listContoh.copy()
newList2[2] = 'baru'

print(newList1)  #  ['hello', 'test', 'coba', 3, True]
print(newList2)  #  ['hello', 'test', 'baru', 3, True]
print(listContoh)  #  ['hello', 'test', 'coba', 3, True]
```

List Concatenating

```
# index =>      0/-5      1/-4      2/-3    3/-2    4/-1
listContoh  =  ['hello',   'coba',   'coba',    3,   True]
listBaru  =  [5, 'test', False]
listCoba  =  ['tarik', 8]

listGabungan = listBaru + listContoh
print(listGabungan)  #  [5, 'test', False, 'hello', 'coba', 'coba', 3, True]
listGabungan.extend(listCoba)
print(listGabungan)  #  [5, 'test', False, 'hello', 'coba', 'coba', 3, True, 'tarik', 8]
```

Find Index

```
# index =>      0/-5      1/-4      2/-3      3/-2      4/-1
listContoh  =  ['eddie',  'baron',   'andi',   'charlie',  'samson']
indexYgDicari1 = listContoh.index('andi')


print(indexYgDicari1)  #  2

indexYgDicari2 = listContoh.index('bernard')   # error
print(indexYgDicari2)
```

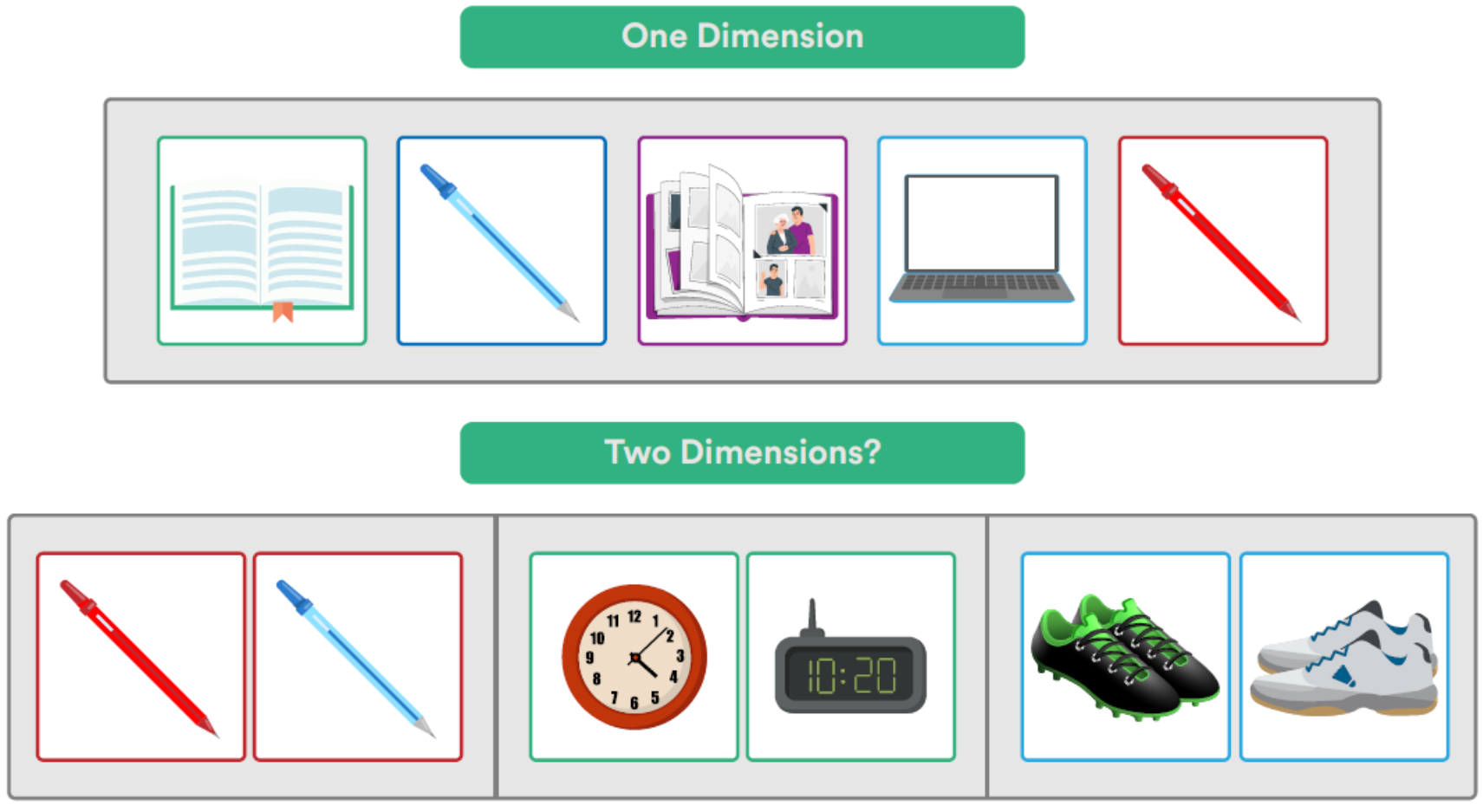
Sorting

```
# index =>      0/-5      1/-4      2/-3      3/-2      4/-1
listContoh  =  ['eddie',  'baron',   'andi',   'charlie',  'samson']
listContoh.sort( )

print(listContoh)  #  ['andi', 'baron', 'charlie', 'eddie', 'samson']
```

Reverse

```
# index =>      0/-5      1/-4      2/-3      3/-2      4/-1
listContoh  =  ['eddie',  'baron',   'andi',   'charlie',  'samson']
listContoh.reverse( )

print(listContoh)  #  ['samson', 'charlie', 'andi', 'baron', 'eddie']
```

2D List



2D List

```
things = [
    ‘red pen’,  ‘blue pen’],
    ‘analog clock’,  ‘digital clock’],
    ‘futsal shoes’,  ‘badminton shoes’]
]
```

```
things[2] = [“futsal shoes”, “badminton shoes”]
things[2][1] = “badminton shoes”
```

Tuple

Tuple Introduction

- What is Tuple?
- Why?
- When?

Syntax Tuple

Syntax Tuple

```
# index =>        0       1   2   3   4
tupleContoh  =  ('hello', 1,  1,  3,  True)
```

Access Data

```
# index =>          0       1   2   3   4
tupleContoh  =  ('hello', 1,  1,  3,  True)

print(tupleContoh[0]) #  hello
print(tupleContoh[2]) #  1
print(tupleContoh[4]) #  True
```

Access Data Negative Indexing

```
# index =>          -5      -4  -3  -2  -1
tupleContoh  =  ('hello',  1,  1,  3, True)

print(tupleContoh[-1])    #  True
print(tupleContoh[-2])    #  3
print(tupleContoh[-5])    #  hello
```

Access Data Range of Indexes

```
# index =>        0/-5      1/-4    2/-3    3/-2    4/-1
tupleContoh  =  ('hello',     1,      1,      3,   True)

print(tupleContoh[-4 : -1])  # (1, 1, 3)
print(tupleContoh[1 : 4])  #  (1, 1, 3)
print(tupleContoh[-5 : 2])  #  ('hello', 1)
print(tupleContoh[0 : -2])  #  ('hello', 1, 1)
print(tupleContoh[2 : ])  #  (1, 3, True)
```

Change Data

Error

```
# index =>        0/-5     1/-4     2/-3     3/-2     4/-1
tupleContoh  =  ('hello',     1,       1,       3,   True)

tupleContoh[1]  =  50  #  TypeError: 'tuple' object does not support item assignment
```

No Error

```
# index =>        0/-5     1/-4     2/-3     3/-2     4/-1
tupleContoh  =  ('hello',     1,       1,       3,   True)

listContoh  = list(tupleContoh)
listContoh[1]  =  50
tupleContoh  =  tuple(listContoh)
print(tupleContoh)  #  ('hello', 50, 1, 3, True)
```

Loop Through a Tuple

```
# index =>        0/-5      1/-4    2/-3    3/-2    4/-1
tupleContoh  =  ('hello',     1,      1,      3,   True)

for item in tupleContoh :
    print(item)

# hello
# 1
# 1
# 3
# True
```

Check if an Item Exists in a Tuple

```
# index =>        0/-5      1/-4    2/-3    3/-2    4/-1
tupleContoh  =  ('hello',     1,      1,      3,   True)

if 'hello' in tupleContoh :
    print("Ada string hello di tupleContoh")
else :
    print("string hello tidak ditemukan di tupleContoh")

# Ada string hello di tupleContoh
```

Length of Tuple

```
# index =>        0/-5      1/-4    2/-3    3/-2    4/-1
tupleContoh  =  ('hello',     1,      1,      3,   True)

print(len(tupleContoh))  #  5
```

Create Tuple with One Item

```
tuple1 = ('hello',)
print(tuple1) # ('hello',)
print(len(tuple1)) # 1
print(type(tuple1)) # <class 'tuple'>

# cara salah/bukan tuple
tuple2 = ('hello')
print(tuple2) # 'hello'
print(type(tuple2)) # <class 'str'>
```

Tuple Concatenating

```
# index =>        0/-5       1/-4       2/-3    3/-2    4/-1
tupleContoh  =  ('hello',     1,         1,      3,    True)
tupleCoba  =  (5, 'test',  False)

tupleGabungan = tupleContoh + tupleCoba
print(tupleGabungan)  #  ('hello', 1, 1, 3, True, 5, 'test', False)
```

Find Index

```
# index =>          0/-5        1/-4        2/-3        3/-2        4/-1
tupleContoh  =  ('eddie',   'baron',    'andi',   'charlie',  'samson')
indexYgDicari1 = tupleContoh.index('andi')

print(indexYgDicari1)  #  2

indexYgDicari2 = tupleContoh.index('bernard')  # error
print(indexYgDicari2)
```

Tuple inside Tuple

```
things = (
    ('red pen', 'blue pen'),
    ('analog clock', 'digital clock'),
    ('futsal shoes', 'badminton shoes')
)
```

```
things[2] = ("futsal shoes", "badminton shoes")
things[2][1] = "badminton shoes"
```

Exercise

Google Search Console
Couverture > Introuvable (404)
Vue d'ensemble
Inspection de l'URL
Performances
Résultats de recherche
Discover
Index
Couverture
Sitemaps
Suppressions
Améliorations
Signaux Web essentiels
Ergonomie mobile
Logos
Extraits d'avis
Toutes les pages connues
Introuvable (404)
État : Exclues
EN SAVOIR PLUS
Pages concernées
52
Exemples
URL
MacBook Air

Soal 1

Bentuk 1

```
* * * * *
* * * *
* * *
* *
*
* *
* * *
* * * *
* * * * *
```

Bentuk 2

```
        *
      * * *
    * * * * *
  * * * * * * *
* * * * * * * * *
  * * * * * * *
    * * * * *
      * * *
        *
```

Soal 2

Proses Menentukan Jumlah Buah

Masukkan Jumlah Apel : 5
Masukkan Jumlah Jeruk : 8
Jumlah yang dimasukkan terlalu banyak
Stock Jeruk tinggal : 7
Masukkan Jumlah Jeruk : 9
Jumlah yang dimasukkan terlalu banyak
Stock Jeruk tinggal : 7
Masukkan Jumlah Jeruk : 7
Masukkan Jumlah Anggur : 8
Jumlah yang dimasukkan terlalu banyak
Stock Anggur tinggal : 6
Masukkan Jumlah Anggur : 6

Detail Belanja

Apel : 5 × 10000 = 50000
Jeruk : 7 × 15000 = 105000
Anggur : 6 × 20000 = 120000

Total : 275000

Proses Bayar

Detail Belanja

Apel : 5 × 10000 = 50000
Jeruk : 7 × 15000 = 105000
Anggur : 6 × 20000 = 120000

Total : 275000

Masukkan jumlah uang : 240000
Uang anda kurang sebesar 35000
Masukkan jumlah uang : 280000
Terima kasih

Uang kembali anda : 5000

Soal 3

Find Min and Max Value

Numbers = [41, 5, 1, 3, 89, 32]
Minimum value = 1
Maximum value = 89

Soal 4

Sort Ascending

List Sebelum Di Sort = [41, 5, 1, 3, 89, 32]
List Setelah Di Sort = [1, 3, 5, 32, 41, 89]

Soal 5

Menu 1. Menampilkan Daftar Buah &
Menu 5. Exit Program

```
Selamat Datang di Pasar Buah

List Menu :
1. Menampilkan Daftar Buah
2. Menambah Buah
3. Menghapus Buah
4. Membeli Buah
5. Exit Program
Masukkan angka Menu yang ingin dijalankan : 1
Daftar Buah

Index   | Nama    | Stock   | Harga
0       | Apel    | 20      | 10000
1       | Jeruk   | 15      | 15000
2       | Anggur  | 25      | 20000

Selamat Datang di Pasar Buah

List Menu :
1. Menampilkan Daftar Buah
2. Menambah Buah
3. Menghapus Buah
4. Membeli Buah
5. Exit Program
Masukkan angka Menu yang ingin dijalankan : 5
```

Soal 5

Menu 2. Menambah Buah

```
                Selamat Datang di Pasar Buah

                List Menu :
                1. Menampilkan Daftar Buah
                2. Menambah Buah
                3. Menghapus Buah
                4. Membeli Buah
                5. Exit Program

                Masukkan angka Menu yang ingin dijalankan : 2
Masukkan Nama Buah      : Nanas
Masukkan Stock Buah     : 10
Masukkan Harga Buah     : 25000
Daftar Buah

Index       | Nama        | Stock     | Harga
0           | Apel        | 20        | 10000
1           | Jeruk       | 15        | 15000
2           | Anggur      | 25        | 20000
3           | Nanas       | 10        | 25000

                Selamat Datang di Pasar Buah

                List Menu :
                1. Menampilkan Daftar Buah
                2. Menambah Buah
                3. Menghapus Buah
                4. Membeli Buah
                5. Exit Program

                Masukkan angka Menu yang ingin dijalankan : ■
```

Soal 5

Menu 3. Menghapus Buah

```
                Selamat Datang di Pasar Buah

                List Menu :
                1. Menampilkan Daftar Buah
                2. Menambah Buah
                3. Menghapus Buah
                4. Membeli Buah
                5. Exit Program

                Masukkan angka Menu yang ingin dijalankan : 3
Daftar Buah

Index           | Nama          | Stock       |  Harga
0               | Apel          | 20          |  10000
1               | Jeruk         | 15          |  15000
2               | Anggur        | 25          |  20000
Masukkan index buah yang ingin dihapus : 1
Daftar Buah

Index           | Nama          | Stock       |  Harga
0               | Apel          | 20          |  10000
1               | Anggur        | 25          |  20000

                Selamat Datang di Pasar Buah

                List Menu :
                1. Menampilkan Daftar Buah
                2. Menambah Buah
                3. Menghapus Buah
                4. Membeli Buah
                5. Exit Program

                Masukkan angka Menu yang ingin dijalankan : 
```

Soal 5

Menu 4. Membeli Buah

```
List Menu :
1. Menampilkan Daftar Buah
2. Menambah Buah
3. Menghapus Buah
4. Membeli Buah
5. Exit Program

Masukkan angka Menu yang ingin dijalankan : 4
Daftar Buah

Index   | Nama    | Stock   | Harga
0       | Apel    | 20      | 10000
1       | Jeruk   | 15      | 15000
2       | Anggur  | 25      | 20000
Masukkan index buah yang ingin dibeli : 1
Masukkan jumlah yang ingin dibeli : 16
Stock tidak cukup, stock Jeruk tinggal 15
Isi Cart :
Nama  | Qty  | Harga
Mau beli yang lain? (ya/tidak) = ya
Masukkan index buah yang ingin dibeli : 1
Masukkan jumlah yang ingin dibeli : 5
Isi Cart :
Nama  | Qty  | Harga
Jeruk | 5    | 15000
Mau beli yang lain? (ya/tidak) = ya
Masukkan index buah yang ingin dibeli : 0
Masukkan jumlah yang ingin dibeli : 4
Isi Cart :
Nama  | Qty  | Harga
Jeruk | 5    | 15000
Apel  | 4    | 10000
Mau beli yang lain? (ya/tidak) = tidak
Daftar Belanja :
Nama  | Qty  | Harga  | Total Harga
Jeruk | 5    | 15000  | 75000
Apel  | 4    | 10000  | 40000
Total Yang Harus Dibayar  =  115000
Masukkan jumlah uang : 125000
Terima kasih

Uang kembali anda : 10000
```

Thank You!

PURWADHIKA HEAD OFFICE

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CAMPUS BRANCHES

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